

Question:
A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

A. Behavior therapy
B. Methimazole
C. Fluoxetine
D. Hearing aids

Reasoning + Answer:

<think>
Finding Reasoning Paths :
1. **Hyper activity and behavioral issues**: Consider attention-deficit/hyper activity disorder (ADHD) as a potential diagnosis , as the child exhibits hyper activity , impulsivity , and difficulty sustaining focus .
2. **Short attention span and in attention**: ADHD is a common cause of short attention span and in attention in children .
3. **Social issues and impulsivity**: Impulsivity may contribute to the child's social challenges , such as hitting others and being disruptive .
4. **Family history**: The father's Graves ' disease and the mother's major depressive disorder may suggest a genetic predis position to behavioral or cognitive issues .
5. **Exclusion of other causes**: Rule out other conditions such as intellectual disability , hearing impairment , or neurodevelopmental disorders like autism spectrum disorder (ASD) based on the clinical presentation .

Reasoning Process :
1. **Symptoms Analysis**:
- The child exhibits hyper activity , impulsivity , and difficulty sustaining focus (short attention span).
- He frequently interrupts others , is more active than peers , and has social challenges (hitting , being disruptive , and frequently bothering others).
- These symptoms align with the diagnostic criteria for **attention-deficit/hyper activity disorder (ADHD)** , specifically the hyper active - impulsive subtype .
2. **Social and Behavioral Issues**:
- The child's impulsivity likely contributes to his social difficulties , such as hitting others and being uncooperative during tasks . These behaviors are consistent with ADHD rather than conditions like autism spectrum disorder (ASD) , which often presents with additional challenges in communication and social interaction .
3. **Short Attention Span and Inattention**:
- The mother reports that the child cannot concentrate for more than a few minutes , which is a hallmark symptom of ADHD . This symptom is not typically associated with conditions like intellectual disability or hearing impairment , as the child appears healthy and well-nourished with no structural abnormalities on examination .
4. **Family History**:
- The father's Graves ' disease is unlikely to directly contribute to the child's behavioral symptoms , as Graves ' disease is an autoimmune condition affecting the thyroid and does not typically present with behavioral abnormalities in children .
- The mother's major depressive disorder may suggest a genetic predis position to emotional or behavioral challenges , but it is not directly linked to ADHD .
5. **Exclusion of Other Diagnoses**:
- **Hearing impairment**: The child does not exhibit signs of hearing loss (e.g ., no mention of difficulty hearing or responding to sounds), and his mother does not report concerns about his hearing .
- **Intellectual disability**: The child has met all developmental milestones , and there is no evidence of cognitive delay .
- **Autism spectrum disorder**: While the child has social challenges , there is no mention of restricted interests , repetitive behaviors , or delayed language development , which are hallmark features of ASD .
6. **Treatment Consideration**:
- ADHD is typically treated with a combination of behavioral therapy and pharmacologic interventions (e.g ., stimulants like methyl phen idate or non-stimulants like atomoxetine).
- Behavioral therapy is the first-line treatment for young children with ADHD , as it focuses on teaching coping strategies , improving social skills , and modifying the child's environment to reduce impulsive and hyperactive behaviors .

Conclusion :
Behavioral therapy is the most appropriate next step in treatment for this child's symptoms , as they are consistent with attention-deficit/hyper activity disorder (AD